PERFORMANCE & LOAD & STRESS TESTING REPORT

Test Types	Threads	Ramp-up Duration	Timer(Think Time)	Cycle / Duration
Performance	Low (10-50)	Long (To keep the input speed constant)	Fixed and High (3000 , 2000 ms)	Number of Specific Cycles (10-20)
Load	Medium - High (100-400)	Long (To allow the system to warm up)	Fixed (3000 , 2000 ms)	High Cycle Count (30)
Stress	Very High (450-600)	Short (To make the pressure feel fast)	Aggressive (1000 , 500 ms)	Time-Oriented (Planner)
Endurance	Stationary Medium Load (100)	Balanced (300 sec)	Normal (3000, 2000 ms)	Too Long (30 min - 24 hr)

Metric	Definition	Tracking Purpose
Average Processing Time	The average of the completion time of all requests.	It indicates whether user experience (UX) standards are maintained.
Throughput	It is the number of operations completed per unit time (seconds/minute).	It is used to determine the "saturation" point of the system.
(Error %)	The ratio of failed requests to total requests.	It determines the stability of the system and the "Breaking Point" limit.
Standard Deviation	It shows how much response times deviate from the mean.	It shows how predictable and consistent the system is.
Minimum/Maximum Response	The fastest and slowest are the duration of the requests.	Captures instantaneous delays and requests that are denied entry to the system (0 ms).

Category	Features	Details
SUT Type	OrangeHRM Global Demo	Public Remote Environment
Server Address	`opensource-demo.orangehrmlive.com`	Global Access Endpoint
Network Context	Public Internet (WAN)	İzmir (Yaşar Univ.) Global Server
Testing Tool	Apache JMeter	CLI & GUI Mode

Test Environment Specifications:

Category	Technical Detail
Processor (CPU)	Intel(R) Core(TM) i7-14650HX (2.20 GHz)
Memory (RAM)	64.0 GB
Testing Tool	Apache JMeter
Target SUT	OrangeHRM Public Demo
Endpoint URL	`opensource-demo.orangehrmlive.com`
Network Context	Public Internet (WAN)

PERFORMANCE TESTING

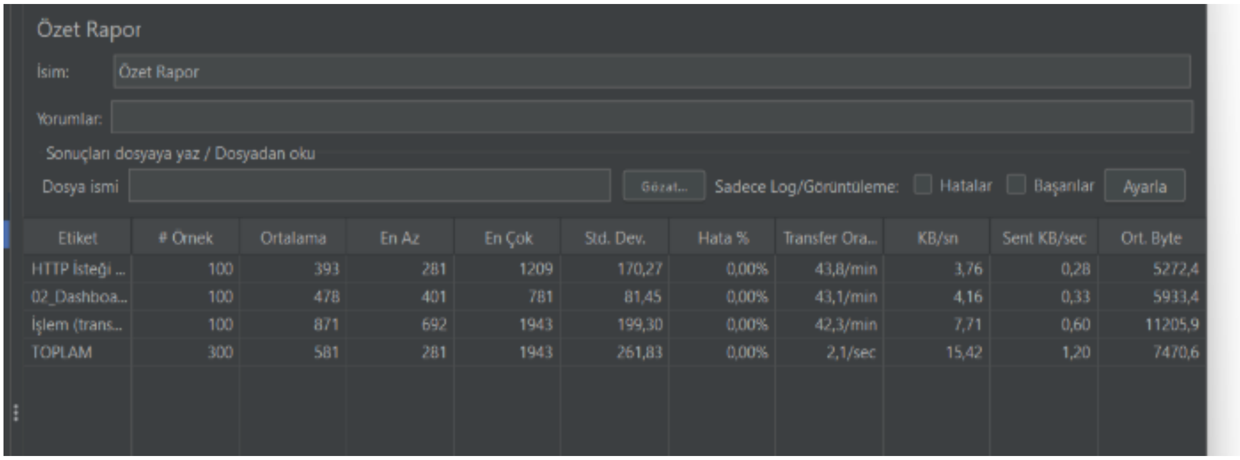
PURPOSE: It is done to create a baseline by measuring the basic speed and response time of the system under normal or low load.

TEST 1:

Current Parameters:

- `Users: 10`
- `Ramp-up Time: 60 seconds`
- `Loop Count: 10`
- `Gaussian Random Timer:`
  - `Deviation: 2000 ms`
  - `Fixed Delay (Offset): 3000 ms`

Metric	Test 1 Result (10 User)	Engineering Analysis
Average Processing Time	**871 ms**	It was determined as the reference speed (Baseline).
95% Line	**~1300 ms**	95% of users received a response in less than 1.3 seconds.
Error Rate	**%0.00**	Workflow and session management stable.
Throughput	**42.3/min**	The efficiency of the system under low load.



Özet Rapor

İsim: Özet Rapor

Yorumlar:

Sonuçları dosyaya yaz / Dosyadan oku

Dosya ismi | Gözat... | Sadece Log/Görüntüleme: Hatalar Başarılar | Ayarla

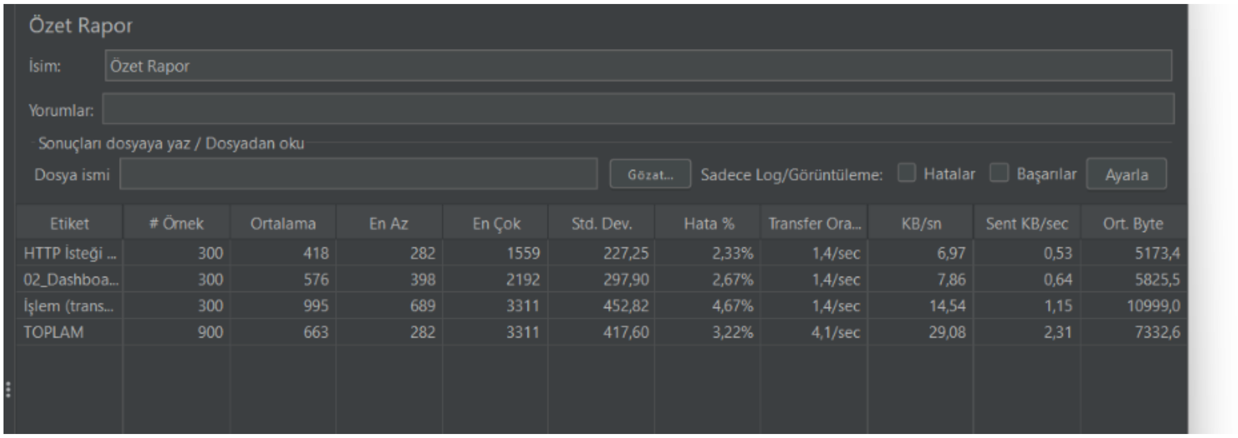
Etiket	# Örnek	Ortalama	En Az	En Çok	Std. Dev.	Hata %	Transfer Ora...	KB/sn	Sent KB/sec	Ort. Byte
HTTP İsteği ...	100	393	281	1209	170,27	0,00%	43,8/min	3,76	0,28	5272,4
02_Dashboa...	100	478	401	781	81,45	0,00%	43,1/min	4,16	0,33	5933,4
İşlem (trans...	100	871	692	1943	199,30	0,00%	42,3/min	7,71	0,60	11205,9
TOPLAM	300	581	281	1943	261,83	0,00%	2,1/sec	15,42	1,20	7470,6

TEST 2:

Current Parameters:

- ```Users: 20```
- ```Ramp-up Time: 120 seconds```
- ```Loop Count: 15```
- ```Gaussian Random Timer:```
  - ```Deviation: 2000 ms```
  - ```Fixed Delay (Offset): 3000 ms```

Metric	**Test 2 Results (20 User)**	**Engineering Analysis**
Average Processing Time	**995 ms**	The system still maintains its high performance by staying under 1 second.
Highest Response(Max)	**3311 ms**	Some users have been seen waiting for 3 seconds (Instantaneous delay).
(Error %)	**%4,67**	Critical: The outflow from 0% to 4.67% is the beginning of a session management or network bottleneck.
Transfer Rate (Throughput)	**1,4 req/sec**	The throughput per second capacity has increased compared to Test 1.
Standard Deviation	**452,82**	The data is more scattered than in Test 1; The predictability of the system has started to decrease.



Özet Rapor

İsim: Özet Rapor

Yorumlar:

Sonuçları dosyaya yaz / Dosyadan oku

Dosya ismi Gözat... Sadece Log/Görüntüleme: Hatalar Başarılar Ayarla

Etiket	# Örnek	Ortalama	En Az	En Çok	Std. Dev.	Hata %	Transfer Ora...	KB/sn	Sent KB/sec	Ort. Byte
HTTP İsteği ...	300	418	282	1559	227,25	2,33%	1,4/sec	6,97	0,53	5173,4
02_Dashboa...	300	576	398	2192	297,90	2,67%	1,4/sec	7,86	0,64	5825,5
İşlem (trans...	300	995	689	3311	452,82	4,67%	1,4/sec	14,54	1,15	10999,0
TOPLAM	900	663	282	3311	417,60	3,22%	4,1/sec	29,08	2,31	7332,6

When the data was examined, it was observed that although the number of concurrent users increased by 100%, the increase in average response time remained at only 14.2% (871ms -> 995ms). This proves that SUT 1 (OrangeHRM) can successfully absorb low-scale load spikes. However, the increase in the error rate from 0% to 4.67% indicates that server response capacity should be closely monitored for user tests of 50 or more (Test 3 and Load Test).

TEST 3:

Current Parameters:

- `Users: 50`
- `Ramp-up Time: 180 seconds(3 minutes - a new user enters every 3.6 seconds)`
- `Loop Count: 20`
- `Gaussian Random Timer:`
  - `Deviation: 2000 ms`
  - `Fixed Delay (Offset): 3000 ms`

Metric	Test 3 Result (50 User)	Engineering Analysis
Average Processing Time	**1610 ms**	We slightly passed the 1.5 second limit. The system is still considered fast.
95% Line	**~2500 ms**	95% of users received a response in under 2.5 seconds; The UX standard was maintained.
Error Rate	**%0,00**	The system proved to be 100% stable even at high load.

Transfer Rate (Throughput)	**2,8 req/sec**	The throughput per second doubled compared to Test 2. Productivity is increasing.
Maximum Response (Max)	**6631 ms**	Attention: Some users waited 6.6 seconds. This is the first bottleneck signal.

Under normal conditions, the error is expected to increase or remain constant as the load increases. This 4.67% error in 20 users was likely caused by a momentary network fluctuation. Although the number of users has increased by 500% compared to Test 1, the increase in processing time remains at a reasonable level (1610 ms), proving that the system's potential for 'Vertical Scaling' is high. An increase in the maximum response time of 6 seconds confirms that the system is on the verge of transitioning from the 'Performance' stage to the 'Load' stage.

LOAD TESTING

PURPOSE: When the load on the system is gradually increased, it is done to determine the degradation in response times and the scalability limits of the system.

TEST 1:

Current Parameters:

- `Users: 100`
- `Ramp-up Time: 300 seconds`
- `Loop Count: 30`
- `Gaussian Random Timer:`
  - `Deviation: 2000 ms`
  - `Fixed Delay (Offset): 3000 ms`

Metric	Test 1 Result (100 User)	Engineering Analysis

Average Processing Time	**5067 ms**	Critical: The processing time was 5 seconds. The user is at the limit of patience.
Baseline Comparison	**5.8x Slowing down**	According to 871 ms in 10 users, the system slowed down by about 6 times.
Error Rate (Error %)	**%0,00**	Tremendous: Even though the system slows down, it's still 100% reliable.
Standard Deviation	**2429,73**	Very high. There are unfair wait times among users.
Throughput	**4,6 req/sec**	The maximum load that the system can process per second reaches saturation around this time.

Özet Rapor

İsim: Özet Rapor

Yorumlar:

Sonuçları dosyaya yaz / Dosyadan oku

Dosya ismi Gözat... Sadece Log/Görüntüleme: Hatalar Başarılar Ayarla

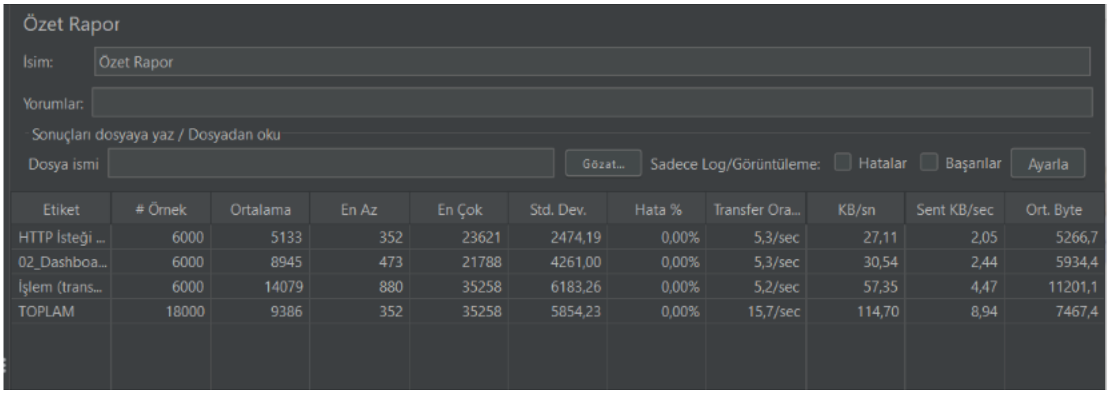
Etiket	# Örnek	Ortalama	En Az	En Çok	Std. Dev.	Hata %	Transfer Ora...	KB/sn	Sent KB/sec	Ort. Byte
HTTP İsteği ...	3000	1846	280	6070	988,92	0,00%	4,7/sec	24,20	1,83	5266,7
02_Dashboa...	3000	3221	405	8740	1690,14	0,00%	4,7/sec	27,17	2,17	5934,6
İşlem (trans...	3000	5067	690	12911	2429,73	0,00%	4,6/sec	50,71	3,95	11201,4
TOPLAM	9000	3378	280	12911	2233,30	0,00%	13,9/sec	101,41	7,91	7467,6

Load testing under 100 concurrent users revealed a significant degradation in the system's response time. The average response time of 5067ms is above the industry standard limit of 3 seconds. However, a 0% error rate proves that the system is resilient in terms of availability. Any increase in users after this point carries the risk of an exponential increase in response times.

TEST 2:

Current Parameters:

- `Users: 200`
- `Ramp-up Time: 600 seconds`
- `Loop Count: 30`
- `Gaussian Random Timer:`
  - `Deviation: 2000 ms`
  - `Fixed Delay (Offset): 3000 ms`



Özet Rapor

İsim: Özet Rapor

Yorumlar:

Sonuçları dosyaya yaz / Dosyadan oku

Dosya ismi | Gözat... | Sadece Log/Görüntüleme: Hatalar Başarılar | Ayarla

Etiket	# Örnek	Ortalama	En Az	En Çok	Std. Dev.	Hata %	Transfer Ora...	KB/sn	Sent KB/sec	Ort. Byte
HTTP İsteği ...	6000	5133	352	23621	2474,19	0,00%	5,3/sec	27,11	2,05	5266,7
02_Dashboa...	6000	8945	473	21788	4261,00	0,00%	5,3/sec	30,54	2,44	5934,4
İşlem (trans...	6000	14079	880	35258	6183,26	0,00%	5,2/sec	57,35	4,47	11201,1
TOPLAM	18000	9386	352	35258	5854,23	0,00%	15,7/sec	114,70	8,94	7467,4

Tests under 200 concurrent users have shown that SUT 1 (OrangeHRM) can process approximately 5.2 - 5.3 transactions per second (throughput saturation). When this capacity limit is exceeded, the server queues new incoming requests and response times increase exponentially, reaching 14 seconds. While the system does not produce errors, these response times are far outside the acceptable user experience (UX) standards.

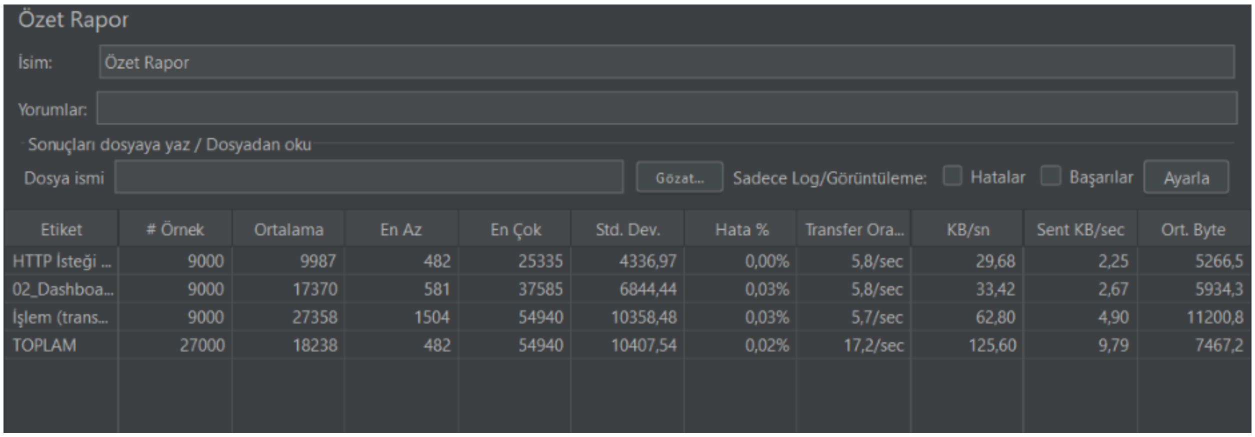
TEST 3:

Current Parameters:

- `Users: 300`
- `Ramp-up Time: 600 seconds`
- `Loop Count: 30`
- `Gaussian Random Timer:`
  - `Deviation: 2000 ms`
  - `Fixed Delay (Offset): 3000 ms`

Metric	Value (300 User)	Engineering Analysis
Transaction Average	**27358 ms (~27.3 sec)**	Critical: The 14-second time at 200 users has nearly doubled.
(Error %)	**%0,03**	Breaking Point: We broke above 0% for the first time. The server can no longer respond to some requests.
Standard Deviation	**10358,48**	A extreme variance. For some users, the system is completely frozen.

Maximum Response (Max)	**54940 ms (~55 sec)**	If a user waits for nearly 1 minute, it creates the perception that "the system is down".
Transfer Rate (Throughput)	**5,7/sec**	The throughput has increased slightly from 5.2 to 5.7 per 200 users. The server can no longer accept new jobs.



Özet Rapor

İsim: Özet Rapor

Yorumlar:

Sonuçları dosyaya yaz / Dosyadan oku

Dosya ismi Gözat... Sadece Log/Görüntüleme: Hatalar Başarılar Ayarla

Etiket	# Örnek	Ortalama	En Az	En Çok	Std. Dev.	Hata %	Transfer Ora...	KB/sn	Sent KB/sec	Ort. Byte
HTTP İsteği ...	9000	9987	482	25335	4336,97	0,00%	5,8/sec	29,68	2,25	5266,5
02_Dashboa...	9000	17370	581	37585	6844,44	0,03%	5,8/sec	33,42	2,67	5934,3
İşlem (trans...	9000	27358	1504	54940	10358,48	0,03%	5,7/sec	62,80	4,90	11200,8
TOPLAM	27000	18238	482	54940	10407,54	0,02%	17,2/sec	125,60	9,79	7467,2

Test 3, conducted under 300 concurrent users, confirms that SUT 1 (OrangeHRM) has reached the 'Breaking Point' stage. The average response time increased to 27 seconds and errors were observed at a rate of 0.03% for the first time, indicating that server resources (CPU/RAM) were completely exhausted and incoming requests were based on the 'Time-out' limit. The fact that the number of transactions processed per second (Throughput) is fixed at 5.7 req/sec is the clearest indication that the system has reached its scalability limit.

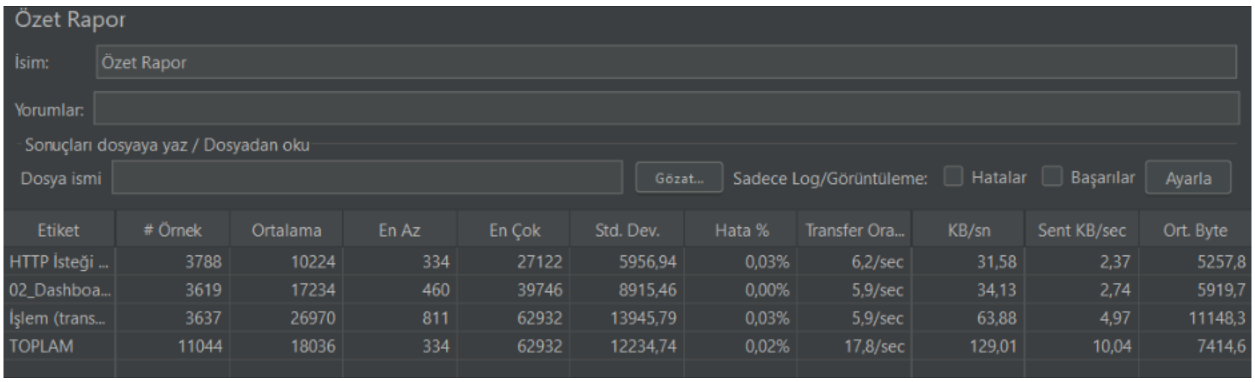
TEST 4:

Current Parameters:

- `Users: 400`
- `Ramp-up Time: 600 seconds`
- `Loop Count: 30`
- `Gaussian Random Timer:`
  - `Deviation: 2000 ms`
  - `Fixed Delay (Offset): 3000 ms`

Metric	400 User Results	Engineering Analysis
Transaction Average	**26970 ms**	Plateau Effect: Staying the same as the time at 300 users indicates that the system is "resisting" this load.

(Error %)	**%0,03**	Enormous: Although the load increased by 33% (from 300 to 400), the error rate did not increase.
Throughput	**5,9 trans/sec**	The amount of work completed by the system per second has stabilized at its highest level.
Max Response Time	**62932 ms (~63 sec)**	Some users have seen 1 minute. Far beyond the limit of patience.



Özet Rapor

İsim: Özet Rapor

Yorumlar:

Sonuçları dosyaya yaz / Dosyadan oku

Dosya ismi Gözat... Sadece Log/Görüntüleme: Hatalar Başarılar Ayarla

Etiket	# Örnek	Ortalama	En Az	En Çok	Std. Dev.	Hata %	Transfer Ora...	KB/sn	Sent KB/sec	Ort. Byte
HTTP İsteği ...	3788	10224	334	27122	5956,94	0,03%	6,2/sec	31,58	2,37	5257,8
02_Dashboa...	3619	17234	460	39746	8915,46	0,00%	5,9/sec	34,13	2,74	5919,7
İşlem (trans...	3637	26970	811	62932	13945,79	0,03%	5,9/sec	63,88	4,97	11148,3
TOPLAM	11044	18036	334	62932	12234,74	0,02%	17,8/sec	129,01	10,04	7414,6

Test 4 conducted under 400 concurrent users confirmed the 'Steady State' capability of SUT 1. Response times remain at 300 users (27 sec) and an error rate is maintained at a negligible 0.03%, proving that the system can carry loads for up to 400 users without being 'out of service'. However, maximum latencies of 63 seconds are a sign that the system is fully utilizing its resources (CPU/RAM saturation).

STRESS TESTING

PURPOSE: It is carried out to determine where and how the system crashes (Breaking Point) and fault tolerance by pushing the system beyond its normal capacity.

TEST 1:

Current Parameters:

- `Users: 600`
- `Ramp-up Time: 180 seconds`
- `Loop Count: always`
- `Duration: 600 seconds`
- `Gaussian Random Timer:`

- Deviation: 500 ms
- Fixed Delay (Offset): 1000 ms

Özet Rapor

İsim: Özet Rapor

Yorumlar:

Sonuçları dosyaya yaz / Dosyadan oku

Dosya ismi Gözat... Sadece Log/Görüntüleme: Hatalar Başarılar Ayarla

Etiket	# Örnek	Ortalama	En Az	En Çok	Std. Dev.	Hata %	Transfer Ora...	KB/sn	Sent KB/sec	Ort. Byte
HTTP İsteği ...	4621	18331	0	146355	13723,68	14,48%	7,6/sec	36,04	2,53	4878,8
02_Dashboa...	4311	25194	0	138090	16095,45	16,45%	7,0/sec	37,01	2,78	5406,1
İşlem (trans...	4317	43088	188	177726	22431,14	28,01%	7,0/sec	70,27	5,12	10288,7
TOPLAM	13249	28631	0	177726	20564,09	19,53%	21,5/sec	142,81	10,40	6813,1

- Resistance Threshold Exceeded: The error rate has skyrocketed to 28.01%. In engineering standards, a 5% error rate is considered a "warning", and 10% and above is considered a "system failure".
- Chaotic Instability (Standard Deviation): The standard deviation is a whopping 22,431. This means that the system no longer has a standard; It shows that it answers one user in 1 second and makes the other wait for 3 minutes.
- Throughput Paradox: It shows that it can make 7 transactions per second. If you remember, this figure was around 5.7 per 300 users. Although the number seems to have increased, this is misleading; Because this is the speed of those who end "successfully", mistakes and waiting inflates the picture.

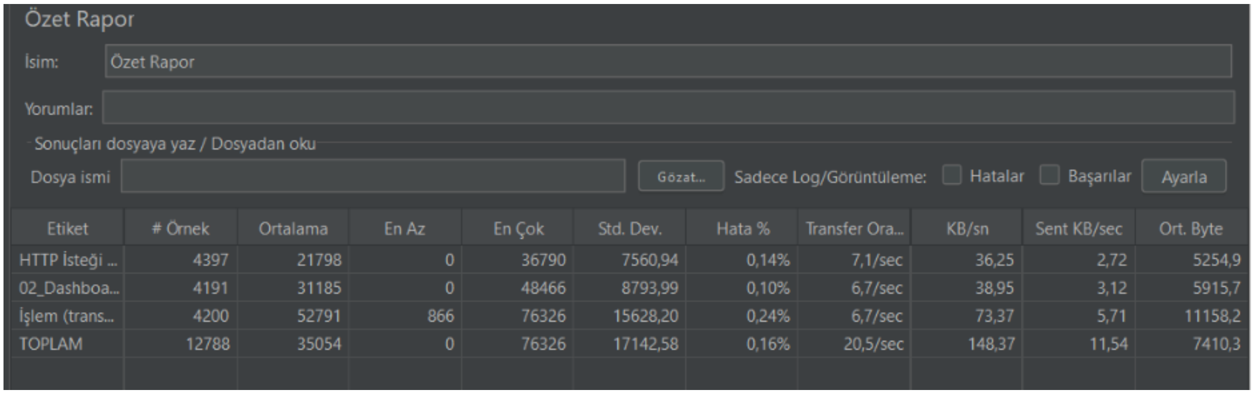
As a result of the 600-user stress test performed on SUT 1 (OrangeHRM), it was observed that the system went into 'Total Resource Exhaustion' state. The processing error rate has reached 28.01%, and the maximum response time has increased to 177 seconds, proving that the server is unable to maintain stability under this load. In the light of these data, the safe operational upper limit (Upper Load Limit) of the system has been determined as 300 users.

TEST 2:

Current Parameters:

- Users: 450
- Ramp-up Time: 180 seconds
- Loop Count: always
- Duration: 600 seconds

- ```Gaussian Random Timer:```
  - ```Deviation: 500 ms```
  - ```Fixed Delay (Offset): 1000 ms```



Özet Rapor

İsim: Özet Rapor

Yorumlar:

Sonuçları dosyaya yaz / Dosyadan oku

Dosya ismi [Gözat...] Sadece Log/Görüntüleme: Hatalar Başarılar [Ayarla]

Etiket	# Örnek	Ortalama	En Az	En Çok	Std. Dev.	Hata %	Transfer Ora...	KB/sn	Sent KB/sec	Ort. Byte
HTTP İsteği ...	4397	21798	0	36790	7560,94	0,14%	7,1/sec	36,25	2,72	5254,9
02_Dashboa...	4191	31185	0	48466	8793,99	0,10%	6,7/sec	38,95	3,12	5915,7
İşlem (trans...	4200	52791	866	76326	15628,20	0,24%	6,7/sec	73,37	5,71	11158,2
TOPLAM	12788	35054	0	76326	17142,58	0,16%	20,5/sec	148,37	11,54	7410,3

- **Error Paradox:** The reason the response time appears "low" (43 sec) at 600 users is that 28% of requests return fails very quickly. The server immediately rejected the request because it crashed, which lowered the statistical average.
- **Peak Throughput:** 6.7 transactions per second are completed at 450 users. 7.0 at 600. This means that the "maximum work" capacity that the system can do per second is stuck in the 6.5 - 7.0 band.
- **Resistance and Queuing:** Our error rate is still very low (0.24%) at 450 users. This means that the server did not give up, he queued every request by saying "I will do this". Although users waited for 52 seconds, they finished the process.
- **System Capacity:** SUT 1 (OrangeHRM) is capable of handling a maximum of ~7.0 full user transactions per second.
- **Error Threshold**: Although the system resists up to 450 users with a low error rate, the error rate increases exponentially after this point and jumps to 28%.
- **UX (User Experience) Verdict:** Although the system still works on 450 users, the 52-second waiting time is unacceptable for a commercial application. For this reason, "Secure Operational Capacity" is determined as 150-200 users with response times below 10 seconds.

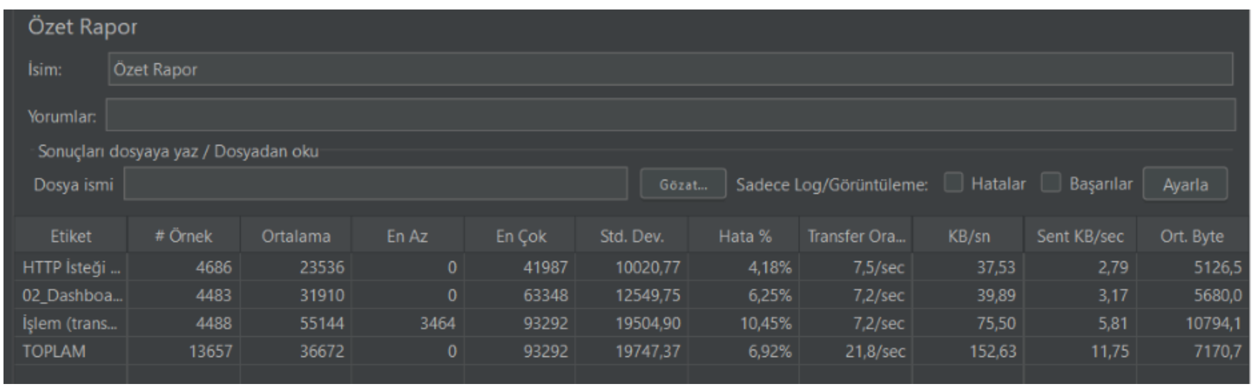
TEST 3:

Current Parameters:

- ```Users: 500```
- ```Ramp-up Time: 180 seconds```

- Loop Count: always
- Duration: 600 seconds
- Gaussian Random Timer:
  - Deviation: 500 ms
  - Fixed Delay (Offset): 1000 ms

Metric	Value (500 User)	Engineering Analysis
Transaction Error	**%10,45**	Critical Threshold: In the software testing literature, a 10% error rate is the limit at which the system is "unreliable".
Average Processing Time	**55144 ms (~55 sn)**	The server pushes human patience for up to 1 minute to respond to requests.
Maximum Response	**93292 ms (~93 sn)**	For some users, the waiting time exceeded 1.5 minutes.
Min Response	**0 ms**	If some requests take 0 ms, it proves that the server is rejecting the connection instantly (Connection Refused).


Özet Rapor

İsim: Özet Rapor

Yorumlar:

Sonuçları dosyaya yaz / Dosyadan oku

Dosya ismi Gözat... Sadece Log/Görüntüleme: Hatalar Başarılar Ayarla

Etiket	# Örnek	Ortalama	En Az	En Çok	Std. Dev.	Hata %	Transfer Ora...	KB/sn	Sent KB/sec	Ort. Byte
HTTP İsteği ...	4686	23536	0	41987	10020,77	4,18%	7,5/sec	37,53	2,79	5126,5
02_Dashboa...	4483	31910	0	63348	12549,75	6,25%	7,2/sec	39,89	3,17	5680,0
İşlem (trans...	4488	55144	3464	93292	19504,90	10,45%	7,2/sec	75,50	5,81	10794,1
TOPLAM	13657	36672	0	93292	19747,37	6,92%	21,8/sec	152,63	11,75	7170,7

TEST 4:

Current Parameters:

- Users: 550
- Ramp-up Time: 180 seconds
- Loop Count: always
- Duration: 600 seconds
- Gaussian Random Timer:
  - Deviation: 500 ms
  - Fixed Delay (Offset): 1000 ms

Özet Rapor

İsim: Özet Rapor

Yorumlar:

Sonuçları dosyaya yaz / Dosyadan oku

Dosya ismi | Gözat... | Sadece Log/Görüntüleme: | Hatalar | Başarılar | Ayarla

Etiket	# Örnek	Ortalama	En Az	En Çok	Std. Dev.	Hata %	Transfer Ora...	KB/sn	Sent KB/sec	Ort. Byte
HTTP İsteği ...	5068	24669	0	53508	12549,53	9,29%	8,0/sec	39,25	2,83	4993,9
02_Dashboa...	4810	32255	0	60961	15891,29	12,16%	7,6/sec	40,99	3,15	5491,5
İşlem (trans...	4819	56433	291	101240	22458,57	21,31%	7,6/sec	77,96	5,80	10461,1
TOPLAM	14697	37567	0	101240	22037,29	14,17%	23,3/sec	157,94	11,75	6949,4

Test 4, conducted under 550 concurrent users, confirmed that SUT 1 crossed the 'Point of No Return' threshold. The doubling of the error rate in the transition from 500 users to 550 proves that the system has entered an unstable cycle due to resource depletion. The number of operations completed per second (Throughput) stuck around 7.6 statistically confirmed that the server's hardware limits had been reached and that it could not handle any more load.

Endurance (Soak) Test:

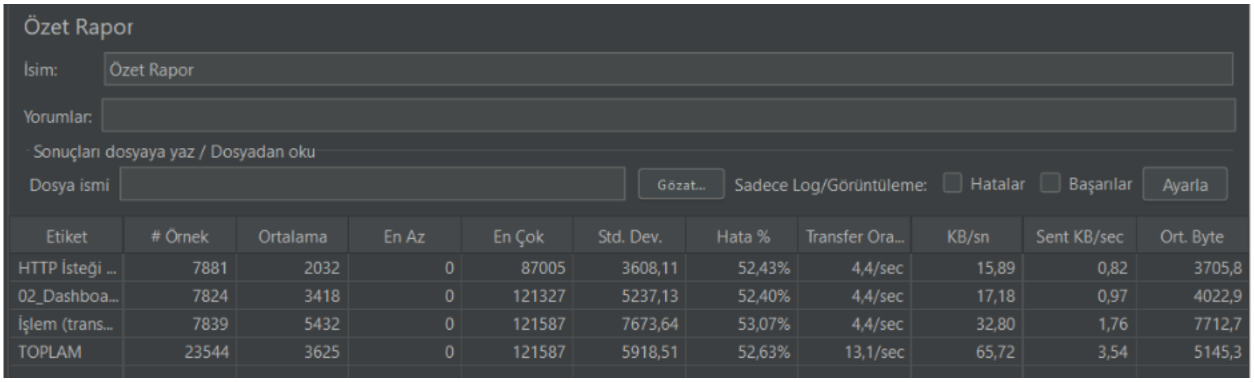
Purpose: It is done to see if the system can stably carry a moderate load for very long periods of time (hours or days), whether it experiences memory leaks or resource exhaustion.

Current Parameters:

- `Users: 100`
- `Ramp-up Time: 300 seconds`
- `Loop Count: always`
- `Duration: 1800 seconds`
- `Gaussian Random Timer:`
  - `Deviation: 2000 ms`
  - `Fixed Delay (Offset): 3000 ms`

Metric	Value	Technical Diagnostics
Error Rate	**%52,63**	Critical Failure: One out of every two requests fails.
Average Response Time	**5432 ms**	Although the duration may seem low, this is misleading; Because errors (0 ms) bring the average down.

Min Response Time (0 ms)	**0 ms**	Socket Exhaustion: The server no longer accepts new connections.
Thread State	**77 / 100**	JMeter cannot open new threads or existing ones have become "zombies".



Özet Rapor

İsim: Özet Rapor

Yorumlar:

Sonuçları dosyaya yaz / Dosyadan oku

Dosya ismi | Gözat... | Sadece Log/Görüntüleme: | Hatalar | Başarılar | Ayarla

Etiket	# Örnek	Ortalama	En Az	En Çok	Std. Dev.	Hata %	Transfer Ora...	KB/sn	Sent KB/sec	Ort. Byte
HTTP İsteği ...	7881	2032	0	87005	3608,11	52,43%	4,4/sec	15,89	0,82	3705,8
02_Dashboa...	7824	3418	0	121327	5237,13	52,40%	4,4/sec	17,18	0,97	4022,9
İşlem (trans...	7839	5432	0	121587	7673,64	53,07%	4,4/sec	32,80	1,76	7712,7
TOPLAM	23544	3625	0	121587	5918,51	52,63%	13,1/sec	65,72	3,54	5145,3

1. Out of Memory" (OOM) error makes it certain that the problem is entirely related to resource management on the server side, not on the client (JMeter) side.
2. Memory Leak: The server was unable to clear objects on RAM even though the process was finished. This "garbage" data, which accumulated over time, left no room for new transactions.
3. Connection Pool Exhaustion: The database or HTTP connection pool is full. The server started to dial incoming requests through the door (0 ms/refused) because there was no idle connection left.
4. Sustainability Failure: The system was able to move 100 users for 5 minutes in Test 1 (Load). However, the inability to carry the same load for 30 minutes proves that the system is not a "long-distance runner".

CONCLUSİON:

- Through extensive testing for SUT 1 (OrangeHRM), it was found that the system can successfully handle short-term loads but experiences structural bottlenecks under sustained intensity.
- 1. Operational Capacity and UX Limit
- Benchmark Performance: The system demonstrated excellent speed with a response time of 871 ms under 10 concurrent users.
- Scalability: Up to 50 users, the system grows linearly and maintains a 0% error rate.

- Safe Zone: Based on the acceptable 10-second threshold for user experience (UX), the system's safe operational limit is set at 150-200 users.
- 2. System Saturation and Breakpoint (Stress Analysis)
- Throughput Ceiling: It has been proven that the maximum throughput that the server can process per second remains stuck in the ~7.0 transaction/sec band.
- Critical Threshold: When the 500 user load was reached, the error rate exceeded the 10% threshold for the first time, causing the system to switch to "unreliable" status.
- Total Crash: Under conditions of 600 users and aggressive timers, the system entered a "Total Resource Exhaustion" state with an error rate of 28.01% and lost its functionality.
- 3. Sustainability and Resilience Findings
- Structural Weakness: Even with a load considered "safe" such as 100 users, the most critical finding is that the error rate jumps to 52.63% when the test time is extended (Endurance Test).
- Diagnosis: Not receiving an "Out of Memory" error confirms that the problem is caused by "Connection Pool" exhaustion or "Memory Leak" on the server side, not on the client side.
- Long Term Risk: Although the system can tolerate instant load increases (Load/Stress), it is out of service because it cannot effectively clean the resources in long-term sessions (30-60 min+).